

Below are the various methods used in NLP to convert words into numbers.

1. One Hot Encoding

One Hot Encoding is a technique used to convert categorical variables into numbers, i.e., they are represented by either a 0 to indicate absence or a 1 to indicate presence. Machines, unlike humans, communicate in the form of 0s and 1s, and thus if the data fed into a computer contains categorical values such as blue, pink, or green, it would not understand how to process them. One Hot Encoding allows the computer to handle these categorical terms by creating separate binary columns, one for each term.

Consider food items such as fries, sushi, and pasta. If we assign them numbers 1, 2, and 3 respectively, the algorithm could misinterpret this by assuming that a lower number implies greater importance. This could lead to inaccurate predictions. Thus, One Hot Encoding allows data to be converted into a format that machine learning models can process. It ensures each category is treated as its own without establishing any false relationships among them.

2. Bag of Words:

Bag of Words is a technique used to process text documents for machine learning models. It basically quantifies the frequency of words occurring in the document and it does so by mapping the points out in a vector space. For example if there are n words then there are n dimensions each dimension representing a word. Then points are marked depending upon what word occurs how many times in the text document, making it easier for the models to understand. It also helps give a jist of what the document might be about depending upon words used and their frequency. If words such as "not satisfied", "service", "poor" occur then it suggests that the document might be a review about a particular service.

However, this also has a disadvantage because two words might have the same spelling but mean different things in different contexts. Some words might come as a pair and do not make sense individually, but here they would be treated as two different words.

This is beneficial in performing simple nlp tasks such as

Text classification- a classic example would be categorising mails as spam and not spam

Sentiment analysis- more enthusiastic words indicate a positive response while the opposite shows the person is unsatisfied/unhappy or a negative response

Document Clustering- if there are multiple documents, the algorithm can divide them into groups depending on their category

3. TF-IDF

In a document there are several common words which occur more than once, such as “the”, “it”, “and”, etc. Now to assess the context of the document if we go by what words occurs the most then it is misleading. Just because a word occurs frequently does not mean it is “important” or determines the context of the document. To ensure this does not happen TF-IDF has been introduced.

TF stands for Term Frequency and IDF stands for Inverse Document Frequency. It is used to determine the importance of a word, which is denoted by its TF-IDF score obtained by multiplying TF and IDF values. A word is said to be important if it occurs the most in one document and the least number of times in the other documents. This shows it's not a common word that occurs everywhere and is more specific.

Term Frequency(TF) means the number of times a word occurs in a given document. Let's say a word occurs 20 times and the total number of words in the document was 500, its TF would be 0.04

Inverse Document Frequency(IDF) means the log to the base 10 of the ratio of total number of documents in the corpus to the number of documents containing the term. A high TF-IDF score is better than a low one as it indicates that the word is of high importance

4. Contextual Embedding

Contextual Embedding is when the model understands the context in which the word is being used rather than just keeping a track of the word. Traditional embeddings assign a vector to each word, and if the words are related to each other such as “maths”, “science”, “subjects” respectively, the vectors assigned to these words would be closer to each other in the embedding space. Although this is helpful, where it might cause an issue in during sentimental analysis or question answering.

If there are two sentences having the same word, Contextual Embedding generates different vectors for the same words depending on their context. By using advanced neural techniques, the model can be trained to understand sentence flow too.

At the main heart of contextual models are LSTM's and Transformers.

LSTM stands for Long-Short Term memory. Here models process each word and as they move forward, they remember the previous word so as to understand the context. Transformers take a different approach where they read the entire sentence in one go which makes understanding the context easier.

All in all contextual embedding is more powerful than traditional embedding and results in more accurate predictions.